



CYCLOIDAL SPEED REDUCERS

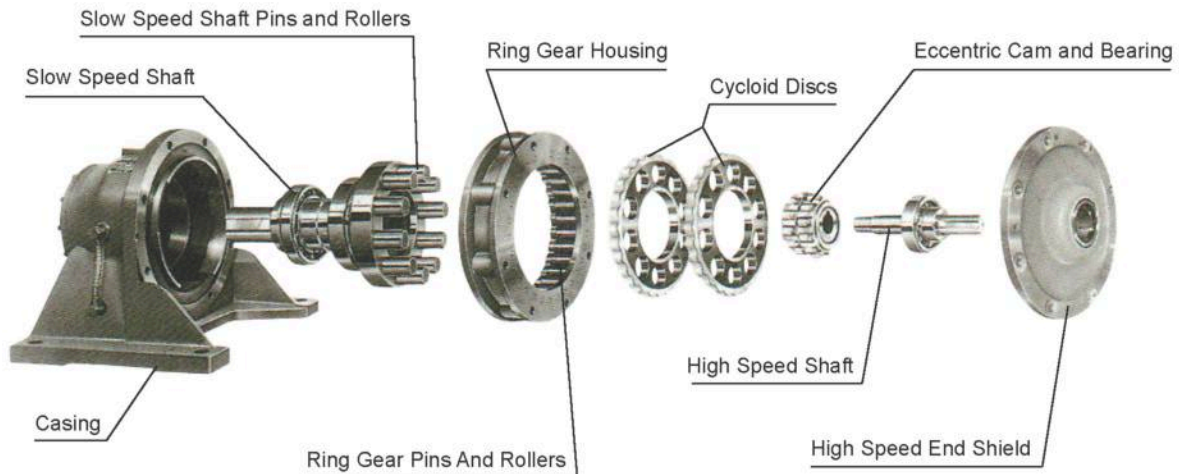


ISO9001:2008RegisteredCompany

GENERAL INFORMATION

The cycloidal design basically has three major components.

- (1) Input shaft assembly (high speed) with eccentric cam, roller bearing and seals.
- (2) Cycloid discs.
- (3) Output shaft assembly (slow speed) with support bearings and seals.



Torque transmitted to the high speed shaft rotates the eccentric cam and roller bearing assembly, and rolls the cycloid discs around the internal circumference of the stationary ring gear housing.

The teeth of the cycloid discs contact the pins of the stationary ring gear, producing a reverse rotation at a reduced speed. Each rotation of the high speed shaft advances the cycloid discs a distance of one tooth pitch in the opposite direction.

The reduced rotation of the cycloid discs is transmitted to the output shaft assembly by means of drive pins and rollers, that are projected through holes located around the bore of the cycloid discs.

Shaft Rotation

For single and triple reduction units, the low speed shaft turns in the opposite direction to the high speed shaft. For double reduction drives, the low speed and high speed shafts turn in the same direction.

Note : On all TRANSCYKO Speed Reducers, the high speed and low speed shafts have the same common axis.

Motor Input Speed (nominal)

1800 R.P.M., 1500 R.P.M., 1200 R.P.M., 1000 R.P.M., 900 R.P.M., 750 R.P.M., 600 R.P.M.

Note : For non-standard input speed, please contact your factory representative to confirm torque ratings.

Shaft Connections

Proper mounting of sheaves, gears, pulleys, chain sprockets, belt and couplings are required to ensure trouble-free operation. All shaft connections must be mounted as close as possible but never touching the cycloidal housing.

Apply overhung loads as close to the bearings as possible to prevent excessive loading on the bearing and possible shaft deflection. It is also strongly suggested that radial loads are applied at the midpoint of the slow speed shaft.

These alignment procedures lead to successful function and efficient performance.

Contact your factory representative to confirm proper selection as needed.

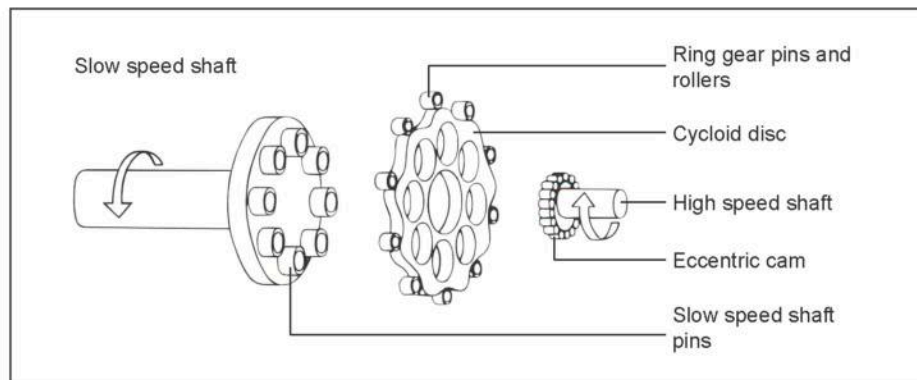
Annotation :

Permissible torsional forces such as thrust and radial loads applied to the low speed shafts must be calculated to determine allowable limits for each selection.

ADVANTAGE OF CYCLO DRIVE

HOW IT WORKS

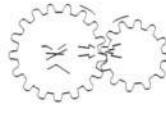
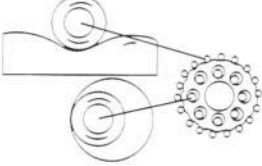
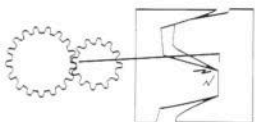
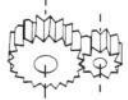
The unique CYCLO Speed reducing system is based on an ingeniously simple principle offering many benefits to the designer and user of power transmission drives. Basically, the speed reducer has only three major moving parts:



- High speed input shaft with integrally mounted eccentric cam and roller bearing assembly
- Cycloid discs,
- Slow speed shaft assembly.

As the eccentric cam rotates, it rolls the cycloid discs around the internal circumference of the stationary ring gear. The resulting action is similar to that of a wheel rolling around the inside of a ring (ring gear housing), the wheel itself turns slowly on its own axis in a counter-clockwise direction. In the CYCLO system the cycloidal profile around the outer edge of the disc engages progressively with the rollers of the fixed ring gear housing to produce a reverse rotation at reduced speed. For each complete revolution of the high speed shaft the cycloid disc turns one cycloidal tooth pitch in the opposite direction. In general, there is one less cycloidal tooth around the disc than there are pins in the fixed ring gear housing, which results in reduction ratios being numerically equal to the number of cycloidal teeth on the disc. (Note: On some ratios, there are two less teeth per cycloid disc than there are pins in the ring gear housing.) The reduced rotation of the cycloid discs is transmitted to the slow speed shaft by means of drive pins and rollers which engage with holes located around the middle of each disc. Normally a two disc system is used with a double eccentric cam which increases the torque capacity and offers an exceptionally smooth vibration-free drive.

ADVANTAGE OF CYCLO-DRIVE

CYCLO-DRIVE	HELICAL GEARS
TWO-THIRDS OF TEETH IN FULL CONTACT ALL TIME  CYCLO SPEED REDUCER Many teeth share the shock of overload.	ONLY 1 OR 2 TEETH IN CONTACT  CONVENTIONAL HELICAL GEAR. 1 or 2 teeth absorb the entire shock of overload.
ROLLING MOTION 	SLIDING FRICTION MOTION 
SHOCK LOAD 500%	SHOCK LOAD ONLY 200%
SMOOTH AND QUIET	NOISY
COMPACT SIZE  COMPACT SIZE Reduction ratios from 6:1 to 119:1 are available for the single stage.	SIZE BIGGER AT THE SAME RATING AND RATIO  Normal ratio of helical gear is 5 or 6

CYCLO - DRIVE GEAR MOTOR THX / THHX

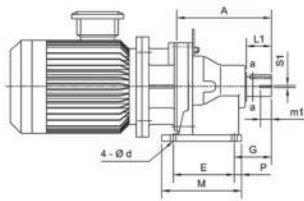


Fig 1. Type THX (0.5~2 HP)
(Grease Lubrication)

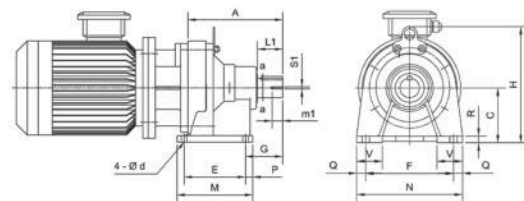


Fig 2. Type THX
(Grease Lubrication)

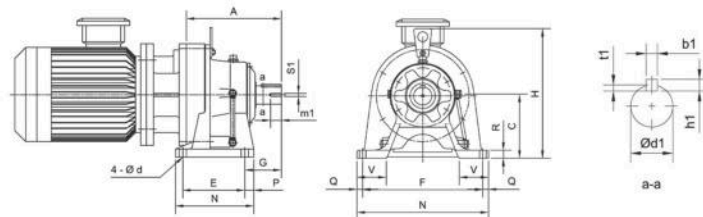
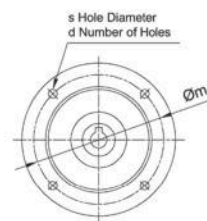
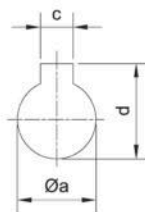
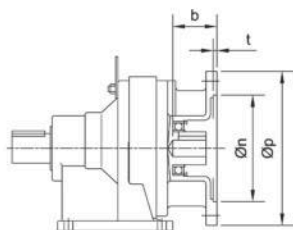


Fig 3. Type THHX
(Oil Lubrication)

Type	A	C	E	F	G	M	N	P	Q	R	V	Slow Speed Shaft					H
												d1	L1	b1	h1	t1	
THX05 - 609 - □	392	100	90	150	60	134	184	17	17	12	40	28	35	8	7	4	207
THX1 - 609 - □	402	100	90	150	60	134	184	17	17	12	40	28	35	8	7	4	207
THX1 - 610 - □	395	100	90	150	60	139	184	17	17	12	40	28	35	8	7	4	207
THX2 - 610 - □	432	100	90	150	60	139	184	17	17	12	40	28	35	8	7	4	207
THX2 - 611 - □	471	120	115	190	82	159	234	22	22	15	55	38	55	10	8	5	257
THX3 - 611 - □	497	120	115	190	82	159	234	22	22	15	55	38	55	10	8	5	257
THX2 - 612 - □	471	140	115	190	82	159	234	22	22	15	55	38	55	10	8	5	277
THX3 - 612 - □	497	140	115	190	82	159	234	22	22	15	55	38	55	10	8	5	277
THX5 - 612 - □	513	140	115	190	82	159	234	22	22	15	55	38	55	10	8	5	280
THX3 - 613 - □	551	150	145	290	100	199	334	27	22	22	65	50	70	14	9	5.5	300
THHX5 - 613 - □	570	150	145	290	100	199	334	27	22	22	65	50	70	14	9	5.5	303
THHX8 - 613 - □	614	150	145	290	100	199	334	27	22	22	65	50	70	14	9	5.5	328
THHX10 - 613 - □	652	150	145	290	100	199	334	27	22	22	65	50	70	14	9	5.5	328
THX3 - 614 - □	571	150	145	290	120	199	334	27	22	22	65	50	90	14	9	5.5	300
THHX5 - 614 - □	590	150	145	290	120	199	334	27	22	22	65	50	90	14	9	5.5	303
THHX8 - 614 - □	634	150	145	290	120	199	334	27	22	22	65	50	90	14	9	5.5	328
THHX10 - 614 - □	672	150	145	290	120	199	334	27	22	22	65	50	90	14	9	5.5	328
THHX8 - 616 - □	675	160	150	370	139	242	414	46	22	25	75	60	90	18	11	7	338
THHX10 - 616 - □	713	160	150	370	139	242	414	46	22	25	75	60	90	18	11	7	338
THHX15 - 616 - □	800	160	150	370	139	242	414	46	22	25	75	60	90	18	11	7	377
THHX15 - 617 - □	844	200	275	380	125	339	434	32	27	30	80	70	90	20	12	7	411
THHX10 - 618 - □	883	220	320	420	145	384	474	32	27	30	85	80	110	22	14	9	446
THHX20 - 618 - □	927	220	320	420	145	384	474	32	27	30	85	80	110	22	14	9	446



Frame No.	m	n	p	t	s	a	b	c	d
63	130	110	160	4.5	9	11	23	4	12.8
71	130	110	160	4.5	9	14	30	5	16.3
80	165	130	200	4.5	11	19	40	6	21.8
90L	165	130	200	4.5	11	24	50	8	27.3
100L, 112M	215	180	250	5	14	28	60	8	31.3
132S, 132S	265	230	300	5	14	38	80	10	41.3
160M, 160L	300	230	350	6	18	42	110	12	45.3

หมายเหตุ : สเป็คมอเตอร์สอบถามรายละเอียดจากผู้จำหน่าย

CYCLO - DRIVE GEAR MOTOR TDVX

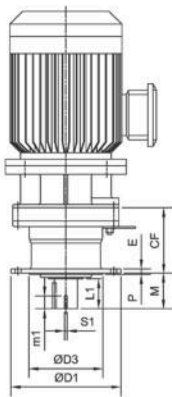


Fig 4.Type TDVX (0.5~2 HP)
(Grease Lubrication)

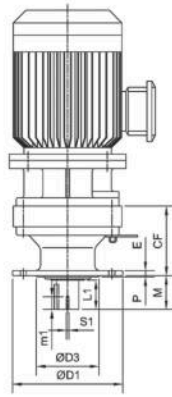


Fig 5. Type TDVX
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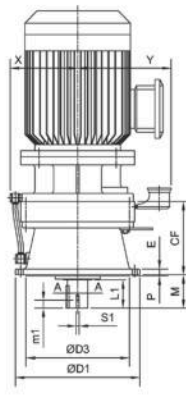


Fig 6. Type TDVX
(Oil Lubrication)

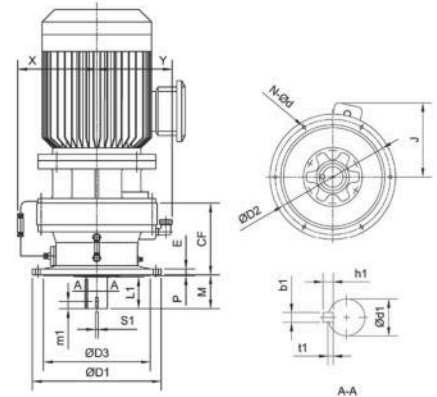
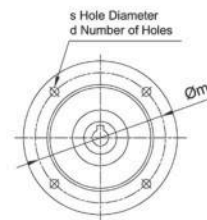
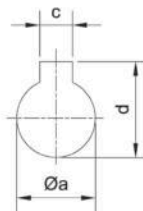
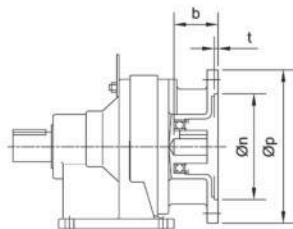


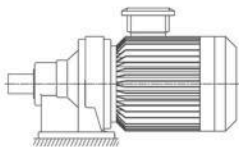
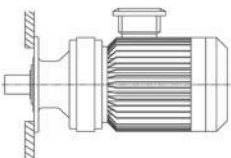
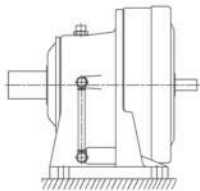
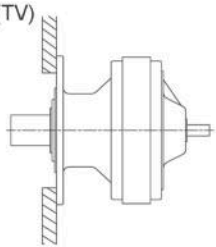
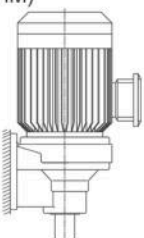
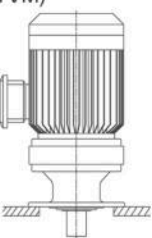
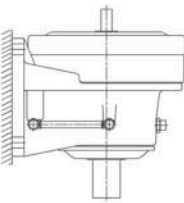
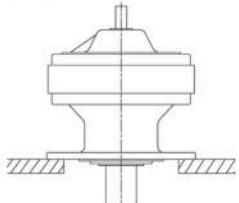
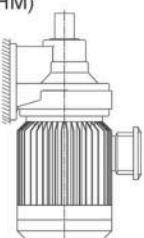
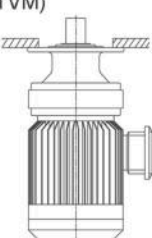
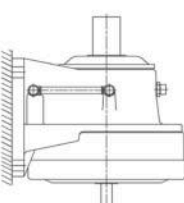
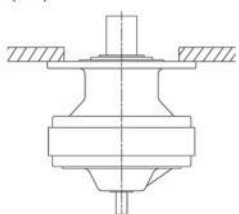
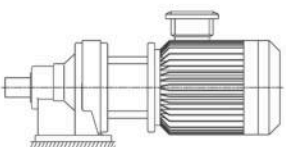
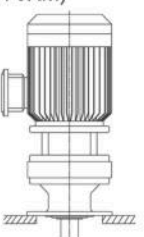
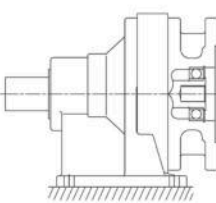
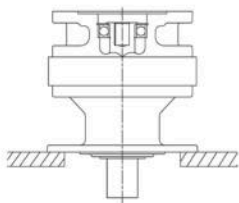
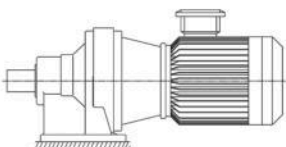
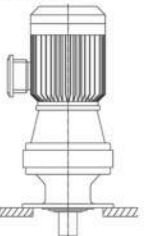
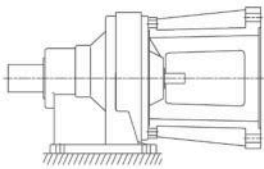
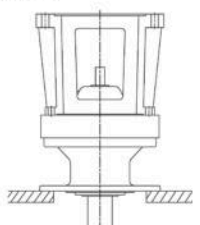
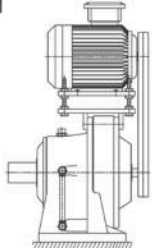
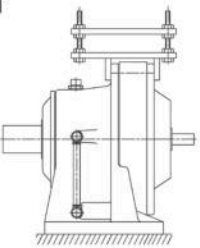
Fig 7.Type TDVX
(Oil Lubrication with circulating pump)

Type	CF	D1	D2	D3	M	E	P	N	d	Slow Speed Shaft				
										d1	L1	b1	h1	t1
TDVX05 - 609 - □	94	160	134	110	48	9	3	4	11	28	35	8	7	4
TDVX1 - 609 - □	94	160	134	110	48	9	3	4	11	28	35	8	7	4
TDVX1 - 610 - □	108	160	134	110	48	9	3	4	11	28	35	8	7	4
TDVX2 - 610 - □	108	160	134	110	48	9	3	4	11	28	35	8	7	4
TDVX2 - 611 - □	117	210	180	140	69	13	4	6	11	38	55	10	8	5
TDVX3 - 611 - □	117	210	180	140	69	13	4	6	11	38	55	10	8	5
TDVX2 - 612 - □	117	210	180	140	69	13	4	6	11	38	55	10	8	5
TDVX3 - 612 - □	117	210	180	140	69	13	4	6	11	38	55	10	8	5
TDVX5 - 612 - □	117	210	180	140	69	13	4	6	11	38	55	10	8	5
TDVX3 - 613 - □	164	260	230	200	76	15	4	6	11	50	70	14	9	5.5
TDVX5 - 613 - □	164	260	230	200	76	15	4	6	11	50	70	14	9	5.5
TDVX8 - 613 - □	164	260	230	200	76	15	4	6	11	50	70	14	9	5.5
TDVX10 - 613 - □	164	260	230	200	76	15	4	6	11	50	70	14	9	5.5
TDVX3 - 614 - □	164	260	230	200	76	15	4	6	11	50	90	14	9	5.5
TDVX5 - 614 - □	164	260	230	200	76	15	4	6	11	50	90	14	9	5.5
TDVX8 - 614 - □	164	260	230	200	76	15	4	6	11	50	90	14	9	5.5
TDVX10 - 614 - □	164	260	230	200	76	15	4	6	11	50	90	14	9	5.5
TDVX8 - 616 - □	219	340	310	270	89	20	4	6	11	60	90	18	11	7
TDVX10 - 616 - □	219	340	310	270	89	20	4	6	11	60	90	18	11	7
TDVX15 - 616 - □	219	340	310	270	89	20	4	6	11	60	90	18	11	7
TDVX15 - 617 - □	258	400	360	316	94	22	5	8	14	70	90	20	12	7
TDVX10 - 618 - □	279	430	390	345	110	22	5	8	18	80	110	22	14	9
TDVX20 - 618 - □	279	430	390	345	110	22	5	8	18	80	110	22	14	9



Frame No.	m	n	p	t	s	a	b	c	d
63	130	110	160	4.5	9	11	23	4	12.8
71	130	110	160	4.5	9	14	30	5	16.3
80	165	130	200	4.5	11	19	40	6	21.8
90L	165	130	200	4.5	11	24	50	8	27.3
100L, 112M	215	180	250	5	14	28	60	8	31.3
132S, 132S	265	230	300	5	14	38	80	10	41.3
160M, 160L	300	230	350	6	18	42	110	12	45.3

MOUNTING POSITIONS

TRANSCYKO Gearmotors		TRANSCYKO Speed Reducers	
THHM(THM) 	THVM(TVM) 	THH(TH) 	THV(TV) 
TDHM(THM) 	TDVM(TVM) 	TDH(TH) 	TDV(TV) 
TUHM(THM) 	TUVM(TVM) 	TUH(TH) 	TUV(TV) 
TUHXM(THXM) 	TDVXM(TVXM) 	THHX(THX) 	TUVX(TVX) 
THHCM(THCM) 	TDVCM(TVCM) 	THHC(THC) 	TDVC(TVC) 
THHTM 		THHT 	

LUBRICATION AND MAINTENANCE

Recommended Grease Lubricants for TRANSCYKO Speed Reducers

Ambient Temperature °C(F)	Maintenanc-Free Grease	Replenish Grease	Electric Motor
-15~50 (5~122)	Shell Alvania Grease RA or equivalent	Shell Alvania Grease No.2, Cosmo Grease Dynamax SH No.2 or equivalent	Shell Alvania Grease No.2 or equivalent

Oil Lubricated Speed Reducers

Oil Lubricated Speed Reducers are factory sealed and pre-lubricated with oil. Please confirm oil level which should be located in between the red lines of oil gauge before operation. Mild EP oil is recommended to lubricate TRANSCYKO Speed Reducers.

Recommended Oil Brand and Specifications(or equivalent)

Ambient Temperature °C (F)	Shell Oil	Mobil Oil	BP Oil	Esso Oil	Gulf Oil	Caltex Oil	SAE Grade	AGMA Grade
-15~5 (5~41)	Omala Oil 86	Mobilgear 626 (ISO VG 68)	Energol GR-xp 46 GR-xp 68	Spartan EP 68	EP Lubricant HD 68	Meropa 68	80W	2EP
0~35 (32~95)	Omala Oil 100, 150	Mobilgear 627 629 (ISO VG 100-150)	Energol GR-xp 100 GR-xp 150	Spartan EP 100 EP 150	EP Lubricant HD 100 HD 150	Meropa 100, 150	80W 90	3EP 4EP
30~50 (86~122)	Omala Oil 220, 320, 460	Mobilgear 630 632, 633, 634 (ISO VG 200-460)	Energol GR-xp 220 GR-xp 320 GR-xp 460	Spartan EP 220 EP 320 EP 460	EP Lubricant HD 220 HD 320 HD 460	Meropa 220, 320, 460	90 140	5EP 7EP

*use lower viscosity oil for winter or relatively low ambient temperature.

There are two types of Forced - Oil Lubrication for vertical units:

- 1.Plunger Pump Lubrication(P) is driven by a cam fitted on the slow speed shaft automatically.
- 2.Trochoid Pump Lubrication(TP):The Oil circulation is conducted by an independent operation of pump and motor.

This table shows the application of Trochoid Pump Lubrication.

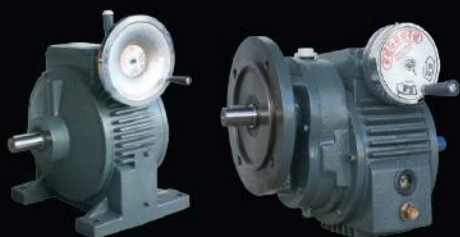
TRANSCYKO Speed Reducers		Trochoid Pump	
Frame Size		Pump Type	Pump Motor
Single Stage	627	TOP-216HA	0.75kW(1 HP), 4p
Double Stage	627/19	TOP-204HA	0.4kW(1/2 HP), 4p

Oil Change Interval

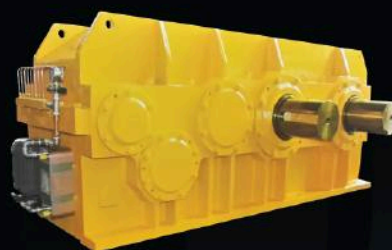
Under all circumstances, each TRANSCYKO unit requires a first oil change post 500 hours of initial operation as a break-in period. Subsequent oil change interval will be based on operation condiiions.

Every 6 months	less than 10 hours / day operation
every 2500 hours	10~24 hours / day operation
every 1~3 months	heavy operation condition such as high ambient temperature & high humidifies.

For specific details of lubrication and maintenance, please refer to the Operating and Maintenance Manual.



Variable Speed Gear Motor



Gear Box



Planetary Gear Box



RV Gear Box For Robot



Helical Bevel Gear



Helical Bevel Gear



High Performance Gear Authorized Distributor



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